



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in

Curriculum for Undergraduate Degree (B.Tech.) in Electronics and Communication (VLSI Design) (w.e.f. AY: 2020-21)

Part III: Detailed Curriculum

Fourth Semester

Course Name:	Digital System Design		
Course Code:	PC-EC402	Category:	Professional Core
Semester:	Fourth	Credit	3
L-T-P:	3-0-0	Prerequisite	Basic Electrical & Electronics Engineering ES-EE101
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	Introduction to basic concept of different number system and their conversions.
2	Developing the concept of venn diagram, Logic gates, SOP, POS and K-Map for logic simplification.
3	Developing the knowledge of different combination and sequential circuits and their design.
4	Developing the conception of different converter circuit and logic families.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	a) Data and number systems; Binary, Octal and Hexadecimal representation and their conversions ; BCD, ASCII, EBCDIC, Gray codes and their conversions; Signed binary number representation with r's and (r-1)'s complement methods.	5
	b) Venn diagram, Boolean algebra; Various Logic gates- their truth tables and circuits; Representation in SOP and POS forms; Minimization of logic expressions by algebraic method, K-map method	6
2	a) Combinational circuits- Adder and Subtractor circuits; Applications and circuits of Encoder, Decoder, Comparator, Multiplexer, De-Multiplexer and Parity Generator, Design of Combinational circuits using MUX,	5
	b) Memory Systems: RAM, ROM, EPROM, E ² PROM	4
	c) Design of combinational circuits-using ROM, Programming logic devices and gate arrays. (PLAs and PLDs)	4



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3	Sequential Circuits- Basic memory element-S-R, J-K, D and T Flip Flops, Race around condition, Various types of Registers and counters , Design of synchronous and asynchronous counter , Irregular counter and design, Introduction to Finite State Machine (FSM) , Design of Sequence Detector using FSM.	6
4	a) Different types of A/D and D/A conversion techniques. b) Logic families- TTL, ECL, Tri-State Logic, MOS and CMOS, their operation and specifications.	4 6
Total		40

Course Outcomes:

After completion of the course, students will be able to:

1	Understanding the different number system and their conversions.
2	Understanding the Venn diagram, logic gates, SOP,POS and K-Map for logic simplification.
3	Design and Analyze the modular combinational circuits
4	Design and Analyze sequential logic circuits.
5	Apply the combinational and sequential circuit understanding for the analyze of converter circuit and logic families.

Learning Resources:

1	Digital Logic and Computer Design, Morries Mano, 2016, Pearson India
2	Modern Digital Electronics, R.P.Jain, 4th ,2010, Mc Graw Hill
3	Digital Electronics: Principles, Devices and Applications, A.K.Maini,2007, Wiley India
4	Fundamentals of Digital Circuits, A.Anand Kumar , 3 rd , PHI
5	Digital Systems Principles and Applications, Ronald J. Tocci , Neal S.Widmer & Gregory L. Moss , 10 th , 2007,Pearson
6	Digital Electronics Principles and Applications ,S.K.Mandal, 2010, Mc Graw Hill.
7	Digital Principles and Application, Leach, Malvino & Saha ,8th, Mc Graw Hill
8	Digital Fundamentals, Floyed & Jain ,8 th ,2005,Pearson

Course Name:	Microprocessors and Microcontrollers		
Course Code:	PC-EC403	Category:	Professional Core
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic Electrical & Electronics Engineering ES-EE101
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05



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Course Objectives:

1	To introduce architecture and operation of microprocessor and microcontroller
2	To learn assembly language programming for microprocessor and microcontroller
3	To understand and design microprocessor and microcontroller based real world applications.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Intel 8085: pin description, architecture, addressing modes, interrupts, timing diagrams. Intel 8086: Pin description, architecture, memory segmentation, pipelining, min/max mode, addressing modes, data structure / access, interrupts.	8
2	Instruction Set and Assembly Language Programming of 8085 and 8086 microprocessors. Instruction formats, addressing modes, instruction set, simple programs involving logical, branch and call instructions, sorting, evaluating arithmetic expressions, string manipulations.	8
3	Memory interfacing of 8086, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine. Introduction to DOS and BIOS interrupts, Interfacing Interrupt Controller 8259, DMA Controller 8257.	4
4	8255 PPI various modes of operation and interfacing to 8085 and 8086. Intel 8279 : Keyboard & display controller ,D/A and A/D converter and other applications.	6
5	Overview of Intel 8051 microcontroller. Architecture. I/O Ports. Memory organization, addressing modes and instruction set of 8051, simple program.	6
6	Serial communication standards, Serial data transfer schemes. 8251 USART architecture and interfacing. RS- 232. IEEE-4-88, Prototyping and trouble shooting, Introduction to Advanced Processors (Intel 80286, Intel 80486) and PIC Microcontroller	8
Total		40

Course Outcomes:

After completion of the course, students will be able to:

1	Apply a basic concept of digital fundamentals to Microprocessor based personal computer system
2	Identify the detailed s/w & h/w structure of the Microprocessor.
3	Illustrate the operation , interface and instructions of microprocessor and microcontroller
4	Apply the microprocessor and microcontroller understanding in a multidisciplinary environment



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Learning Resources:

1	Microprocessor Architecture, Programming and Applications with the 8085, Ramesh Gaonkar, 2013, Penram International Publishing. .
2	Fundamentals of Microprocessor and Microcomputer, B Ram, 2017, Dhanpat Rai Publications.
3	Advanced Microprocessor and Peripherals, K M Bhurchandi, A K Ray, 2017, McGraw Hill Education.
4	The 8051 Microcontroller, Kenneth J. Ayala, 1996, PenramInternational Publishing
5	The 8051 Microcontroller and Embedded Systems: Using Assembly and C, M. A. Mazidi, J. G. Mazidi and R D McKinlay, 2007, Pearson.
6.	Microprocessors & Interfacing, Douglas V. Hall and SSSP Rao, 2017, McGraw Hill Education.
7	Computer Organization and Design: The Hardware/Software Interface, David A. Patterson, John L. Hennessy, 2016, Morgan Kaufmann Publishing

Course Name:	Analog and Digital Communication		
Course Code:	PC-EC(V)401	Category:	Professional Core
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Signals and Systems, PC-EC303
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To develop the basic concept of analog and digital communication techniques.
2	To have knowledge of amplitude modulation and Frequency modulation.
3.	To understand the basic building blocks of super heterodyne receiver and its merits and demerits.
4.	To understand the basic building blocks of digital communication system and its advantages.
5.	To understand the representation of digital data using different modulation techniques, generation and detection of modulated signals.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction to Communication Introduction to Communication, Elements of communication system - Transmitters, Transmission channels & receivers, Concept of Baseband	2



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	signal, Concept of modulation, its needs.	
2	Continuous Wave Linear Modulation: (a) Basic principle of linear (AM) modulation, Time and frequency domain representation of AM signal, Generation of AM waves, Demodulation of AM wave. Double side band suppressed carrier (DSBSC) modulation: time and frequency domain expressions, bandwidth and transmission power for DSB. Single side band modulation (SSB) both TC & SC and only the basic concept of VSB, Spectra and band-width. Calculation of modulation index, power and bandwidth. Basic principle of nonlinear (FM, PM) modulation. Generation of FM waves. Time and Frequency domain representations, Demodulation of FM waves, Calculation of modulation index, power and bandwidth	10
3	Superheterodyne Receiver and Noise: (a) Principle of Super heterodyne receivers: Super heterodyning principle, intermediate frequency, Local oscillator frequency, image frequency. (b) Random Signals and Noise in Communication System: Signal-to-Noise ratio, White noise, thermal noise, Figure of Merit.	2 4
4	Sampling & PCM: Introduction to Digital Communication, Concept of sampling, Pulse Amplitude Modulation (PAM), Pulse Code Modulation (PCM), quantization, uniform and non-uniform quantization, quantization noise, binary encoding, A-Law and μ -law companding, differential PCM, delta modulation and adaptive delta modulation. Digital transmission components, source, multiplexer, line coder, regenerative repeater, Bit rate, Baud rate, M-ary encoding, concept of line coding – polar/unipolar/bipolar NRZ and RZ, Manchester, Inter Symbol Interference (ISI), Eye pattern.	10
5	Digital carrier modulation & demodulation technique: Geometric interpretation of signals, Types of Digital Modulation, coherent and non-coherent Binary Modulation Techniques, Basic digital carrier modulation techniques: -ASK, BPSK, BFSK, QPSK, generation and detection of Signal, power spectrum. Introduction to QAM, Minimum Shift Keying (MSK), constellation diagrams	8
Total		36



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Course Outcomes:	
After completion of the course, students will be able to:	
1	Understand the need for modulation, representation of the modulated carrier wave in both analog and digital communication.
2	Identify the different types of analog and digital modulation technique, identify the modulating and carrier frequencies, and decide the type of generation method to be adopted.
3	Understand the basic concept of Sampling and analog signal to digital signal conversion techniques, concepts of Digital transmission, concept of line coding and ISI
4	Analyze different career modulation techniques considering noise aspects
5	Illustrate the constellation diagram for different digital modulation technique.

Learning Resources:	
1	Modern Digital and Analog Communication Systems, by B. P Lathi. Oxford University Press, 4 th edition, 2011
2	Principles of Communication Systems by H. Taub and D.L.Schilling, Goutam Saha, TMH Publishing Co., 4 th edition, 2017
3	Communication Systems by Simon Haykin, Wiley, 4th edition, 2006
4	Analog and Digital Communications by Dr Sanjay Sharma, Kataria Publishers, 7 th edition, 2017

Course Name	Control System		
Course Code:	PC-EC(V)402	Category:	Professional Core
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic knowledge of mathematics 2 & Network Theory
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To make the students knowledgeable with control system and make them able to illustrate the mathematical representation of the real SISO & MIMO systems.
2	To make the students able to determine the system response to analyze the transient & steady state performance of different systems.
3	To make the students efficient to analyze the stability of the systems using different techniques and be able to design some classical control system.



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Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	a) INTRODUCTION Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems, Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations, Impulse Response and transfer functions –transfer functions of electrical network. Translational and Rotational mechanical systems , Mechanical and electrical analogous systems.	5
	b) TRANSFER FUNCTION REPRESENTATION Transfer Function of linear systems, Block diagram representation of systems considering electrical systems as examples -Block diagram algebra –Representation by Signal flow graph - Reduction using Mason's gain formula.	5
2.	a) TIME RESPONSE ANALYSIS Standard test signals - Time response of first order systems – Characteristic Equation of Feedback control systems, Transient response of second ordersystems - Time domain specifications – Steady state response - Steady state errors and error constants.	4
	b) STABILITY ANALYSIS IN S-DOMAIN The concept of stability – Routh's stability criterion – limitations of Routh's stability. Root Locus Technique: The root locus concept - construction of root loci-effects of adding poles and zeros to $G(s)H(s)$ on the root loci.	4
3.	a) FREQUENCY RESPONSE ANALYSIS Introduction, Frequency domain Specifications-Bode Diagrams- Determination of Frequency domain specifications and transfer function from the Bode Diagram-Phase margin and Gain margin-Stability Analysis from Bode Plots.	5
	b) STABILITY ANALYSIS IN FREQUENCY DOMAIN Polar Plots, Nyquist Plots , Nyquist stability criteria, Stability Analysis.	3
4.	a) CLASSICAL CONTROL DESIGN TECHNIQUES Compensation techniques – Lag, Lead, Lead-Lag. Controllers design in frequency Domain, P, I, D and PID Controllers- basic concepts.	5
	b) STATE SPACE ANALYSIS OF CONTINUOUS SYSTEMS Concepts of state, state variables and state model, derivation of state models from block diagrams, Diagonalization- Solving the Time invariant state Equations- State Transition Matrix and it's Properties – Concepts of Controllability and Observability of a system, Kalman's Test.	5
Total		36



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Course Outcomes:

After completion of the course, students will be able to:

1	State basic knowledge of control system and illustrate mathematical representation of the real SISO & MIMO systems.
2	Determine the system response to analyze the transient & steady state performance of the systems.
3	Perform stability analysis of system using different technique.
4	Design some classical control system.

Learning Resources:

1	Control Systems Engineering – by I. J. Nagrath and M. Gopal, New Age International (P) Limited, Publishers, 2nd edition.
2	Control System Engineering – by P. Ramesh Babu, SCITECH
3	Automatic Control Systems 8th edition– by B. C. Kuo 2003– John Wiley and son's.,
4	Modern Control Engineering – by Katsuhiko Ogata – Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.
5	Linear Control Systems – by B.S. Manke – Khanna Publishers

Course Name:	Python Programming		
Course Code:	OE- EC(V)401A	Category:	Open Elective
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Familiar with any Basic computer language
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	a) Master the fundamentals of writing Python scripts b) Learn core Python scripting elements such as variables and flow control structures
2	a) Use Python to read and write files b) Make their code robust by handling errors and exceptions properly
3	a) Explore Python's object-oriented features b) Search text using regular expressions
4	Understand the Exception Handling and Object oriented concept of Python.



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Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction History, Features, Setting up path, Working with Python, Basic Syntax, Variable and Data Types, Operator	3
2	Conditional Statements If, If- else, Nested if-else, Looping, For, While, Nested loops	3
3	Control Statements Break, Continue, Pass, String Manipulation Accessing Strings, Basic Operations, String slices, Function and Methods	3
4	Lists Introduction, Accessing list, Operations, Working with lists, Function and Methods Tuple Introduction, Accessing tuples, Operations, Working, Functions and Methods Dictionaries Introduction, Accessing values in dictionaries, Working with dictionaries, Properties	6
5	Functions Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables Modules Importing module, Math module, Implementation of Array using numpy module, Random module, Packages, Composition, Input-Output	6
6	Exception Handling Exception, Exception Handling, Except clause, Try- finally clause, User Defined Exceptions	3
7	The Object-Oriented Approach: Classes, Methods, Objects Basic concepts of object programming, Implementation of Properties of Object-oriented Programming, Inheritance.	3
8	File Handling Introduction to File Handling, Data Files, Opening and Closing Files, Reading and Writing Files	3
9	Data Visualization using Python Concept of Data Visualization, Using Pyplot of Matplotlib library, Creating Line chart, Bar chart and Pie chart using pyplot interface, Customizing the Plot	3
Total		33



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Course Outcomes:

After completion of the course, students will be able to:

1	Understand python object types, branching and looping structure in python programming.
2	Use lists, tuples, and dictionaries, functions, packages, concept of files in Python programs
3	Design object oriented programs with Python classes.
4	Use exception handling in Python applications for error handling.

Learning Resources:

1	"Core Python Programming" by R.Nageswar Rao
2	"Python:the complete reference" by Martin C.Brown
3	"Let us Python" by Yashvant Kanetkar
4	"Programming and Problem Solving With Python" by Ashok Namdev Kamthane and Amit Ashok Kamthane
5	"Python Programming" by Anurag Gupta and G. P. Biswas

Course Name:	Object Oriented Programming		
Course Code:	OE-EC(V)401B	Category:	Open Elective
Semester:	Fourth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic Knowledge of Programming
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To understand Object Oriented Design principle and Programming concepts using characteristics of Java
2	To know the principles of Inheritance, Packages and Interfaces
3	To define Exceptions and use I/O streams
4	To develop synchronization applications with Java threads and to develop generics classes
5	To design Graphical User Interfaces through Java
6	To handle Database related problems through JDBC tools

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Object Oriented Design: Concepts of object oriented programming language, Major and minor elements, Object, Class, relationships among objects, aggregation, links,	6



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	relationships among classes-association, aggregation, using, instantiation, meta-class, grouping constructs	
2	Features of object-oriented programming: Objects and Classes, Abstraction, Encapsulation, Inheritance, Polymorphism	4
3	Basic concepts of Object Oriented Programming using Java: Class & Object properties: Basic concepts of java programming – advantages of java, byte-code & JVM, data types, access specifiers, operators, control statements & loops, array, creation of class, object, constructor, finalize and garbage collection, use of method overloading, this keyword, use of objects as parameter & methods returning objects, call by value & call by reference, static variables & methods, garbage collection, nested & inner classes, basic string handling concepts-concept of mutable and immutable string, command line arguments, basics of I/O operations – keyboard input using Buffered Reader & Scanner classes.	7
4	Reusability properties: Super class & subclasses including multilevel hierarchy, process of constructor calling in inheritance, use of super and final keywords with super() method, dynamic method dispatch, use of abstract classes & methods, interfaces. Creation of packages, importing packages, member access for packages	6
5	Exception Handling and Multithreading : Exception handling basics, different types of exception classes, use of try & catch with throw, throws & finally, creation of user defined exception classes. Basics of multithreading, main thread, thread life cycle, creation of multiple threads, thread priorities, thread synchronization, inter-thread communication, deadlocks for threads, suspending & resuming threads	5
6	Generic Programming: Generic classes – generic methods – Bounded Types – Restrictions and Limitations.	3
7	Graphical programming with AWT and Swing: Frame, Components, working with 2D shapes, Using color, fonts, and images, Basics of event handling, event handlers, adapter classes, actions, mouse events, AWT event hierarchy Introduction to Swing, layout management, Swing Components, Text Fields , Text Areas , Button-Check Boxes-Radio Buttons, Lists, choices, Scrollbars, Windows –Menus – Dialog Boxes	4
8	Database Connectivity through JDBC tools	2
Total		37



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Course Outcomes:

After completion of the course, students will be able to:

1	Recognize features of object-oriented design such as encapsulation, polymorphism, inheritance, and composition of systems based on object identity
2	Express basic techniques of writing programs using loops, methods and arrays in object oriented paradigm
3	Apply the concepts of Encapsulation, Inheritance, Polymorphism in developing object oriented featured software.
4	Design applications with an event-driven graphical user interface and using databases
5	Practice efficient mapping of the real world interdisciplinary problems into object oriented programming methodologies.

Learning Resources:

1	Rambaugh, James Michael, Blaha – "Object Oriented Modelling and Design" – Prentice Hall, India
2	Ali Bahrami – "Object Oriented System Development" – Mc Graw Hill
3	Patrick Naughton, Herbert Schildt – "The complete reference-Java2" – TMH
4	Deitel and Deitel – "Java How to Program" – 6th Ed. – Pearson
5	Ivor Horton's Beginning Java 2 SDK – Wrox
6	E. Balagurusamy – "Programming With Java: A Primer" – 3rd Ed. – TMH

Course Name:	Biology		
Course Code:	BS-BIO401	Category:	Basic Science Course
Semester:	Fourth	Credit:	2
L-T-P:	2-0-0	Pre-Requisites:	Basic knowledge of Physics, Chemistry and Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	Bring out the fundamental differences between science and engineering
2	Discuss how biological observations of 18th Century that lead to major discoveries

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Module 1- Introduction to Biology: To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry. Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most	2



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	exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18th Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.	
2	Module2-Classification System in Biology: The underlying criterion, such as morphological, biochemical or ecological be highlighted. Hierarchy of life forms at phenomenological level. A given organism can come under different category based on classification. Model organisms for the study of biology come from different groups. <i>E. coli</i> , <i>S. cerevisiae</i> , <i>D. melanogaster</i> , <i>C. elegance</i> , <i>A. thaliana</i> , <i>M. musculus</i> .	2
3	Module 3: Genetics: To convey that "Genetics is to biology what Newton's laws are to Physical Sciences" Mendel's laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Gene interaction, Epistasis. Meiosis and Mitosis be taught as a part of genetics. Emphasis to be given not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Importance of stem cell research.	2
4	Module 4: Biomolecules: To convey that all forms of life have the same building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. In this context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA.	4
5	Module 5: Enzymes: To convey that without catalysis life would not have existed on earth Enzymology: How to monitor enzyme catalysed reactions. How does an enzyme catalyse reactions? Discuss at least two examples.	2
6	Module 6: Information Transfer: The molecular basis of coding and decoding genetic information is universal Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA structure- from single stranded to double helix to nucleosomes. Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination.	4
7	Module 7: Macromolecular analysis: How to analyse biological processes at the reductionist level Proteins-structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Proteins as enzymes, transporters, receptors and structural elements.	4



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8	Module 8: Metabolism: ATP as an energy currency. This should include the breakdown of glucose to CO ₂ + H ₂ O (Glycolysis and Krebs cycle) and synthesis of glucose from CO ₂ and H ₂ O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge.	2
9	Module 9: Microbiology: Concept of microscopic organisms. Concept of species and strains. Identification and classification of microorganisms. Sterilization and media compositions. Growth kinetics. Microscopy: simple, compound, phase-contrast, SEM, TEM, Confocal: principle and applications.	2
Total		24

Course Outcomes:

After completion of the course, students will be able to:

1	State different engineering applications from biological perspective.
2	Classify biological systems and identify different organisms and microorganisms depending on their morphological, biochemical and ecological criterion.
3	Explain the concept of recessiveness and dominance during the passage of genetic material from parent to offspring and describe DNA as a genetic material in the molecular basis of information transfer.
4	Discuss structures of different biomolecules starting from basic units and hence understand different biological processes at the reductionistic level.
5	Describe protein structures and enzymology and also compare different mechanisms of enzyme action.
6	Describe energy transformation processes in biological systems.

Learning Resources:

1	Biology for Engineers. Arthur T. Johnson. CRC Press.
2	Biology and Engineering of Stem Cell Niches. A K Vishwakarma and Jefferey Karp, Elsevier.
3	Environmental Biology for Engineers and Scientists. David A. Vaccari, P. P. Storm and J. F Alleman. ELBS
4	Biology for Engineers. G. K. Suraishkumar. Oxford



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Course Name:	Digital System Design Lab		
Course Code:	PC-EC492	Category:	Professional Core
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Basic Electrical & electronics Laboratory
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	Introduction to digital lab related all required circuit components.
2	Developing concept of different combinational circuit making and verifying its operations.
3	Developing concept of different sequential circuit making and verifying its operations.

Course Contents:		
Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Realization of basic gates using Universal logic gates.	2
2	Implementation of the Given Boolean Function using Logic Gates in Both SOP and POS Forms.	2
3	Realization of suitable code conversion circuits and vice-versa.	2
4	Realization of parity generator and comparator circuits.	2
5	Construction of simple Encoder and Multiplexer circuits using logic gates.	2
6	Construction of simple Decoder and De-Multiplexer circuits using logic gates.	2
7	Construction of simple arithmetic circuits-Adder, Subtractor.	2
8	Realization of RS,JK,T and D flip-flops using suitable logic gates.	2
9	Realization of Asynchronous Up/Down counters.	2
10	Realization of Synchronous Up/Down counters.	2
11	Design of Sequential Counter with irregular sequences.	2
12	Realization of Ring counter and Johnson's counter.	2
Total		24

Course Outcomes:	
After completion of the course, students will be able to:	
1	Identify the various digital ICs and understand their operation
2	Develop the concept of Boolean laws and digital systems.
3	Apply Boolean laws for the different combinational logic circuit design.
4	Design the different sequential logic circuits.



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Learning Resources:	
1	Digital Logic and Computer Design, Morries Mano, 2016, Pearson India
2	Modern Digital Electronics, R.P.Jain, 4th ,2010, Mc Graw Hill
3	Digital Electronics: Principles, Devices and Applications, A.K.Maini,2007, Wiley India
4	Fundamentals of Digital Circuits, A.Anand Kumar , 3 rd , PHI
5	Digital Systems Principles and Applications, Ronald J. Tocci , Neal S.Widmer & Gregory L. Moss , 10 th , 2007,Pearson
6	Digital Electronics Principles and Applications ,S.K.Mandal, 2010, Mc Graw Hill.
7	Digital Principles and Application, Leach, Malvino & Saha ,8th, Mc Graw Hill
8	Digital Fundamentals, Floyed & Jain ,8 th ,2005,Pearson

Course Name:	Microprocessors and Microcontrollers Lab		
Course Code:	PC-EC493	Category:	Professional Core
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Basic Electrical & Electronics Laboratory
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	To write and execute assembly language program using a trainer kit and simulator
2	To understand interfacing
3	To design and implement microprocessor and microcontroller based applications

Course Contents:		
Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Familiarization with Intel 8085 & Intel 8051simulator on PC	2
2	Study of prewritten programs using basic instruction set (data transfer, Load/Store, Arithmetic, Logical) on the KIT	2
3	Programming using kit and simulator for: i) Table look up ii) Copying a block of memory iii) Shifting a block of memory iv) Packing and unpacking of BCD numbers v) Addition of BCD numbers vi) Binary to ASCII conversion vii) String Matching, Multiplication using shift and add method and Booth's Algorithm	4
4	Program using subroutine calls and IN/OUT instructions using 8255 PPI on the trainer kit e.g. subroutine for delay, reading switch state and	2



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	glowing LEDs accordingly.	
5	Study of timing diagram of an instruction on oscilloscope. Serial communication between two trainer kits.	2
6	Interfacing of 8255: Keyboard and Multi-digit Display with multiplexing using 8255. Study of 8051 Micro controller kit and writing programs as mentioned in S/L3.	6
7.	Writing of programs to interface of Keyboard, DAC and ADC using the kit.	2
Total		20

Course Outcomes:

After completion of the course, students will be able to:

1	Illustrate & Execute assembly language program using kit and simulator
2	Apply the interface of PPI to microprocessor and microcontroller
3	Design and implement applications based on microprocessor and microcontroller

Learning Resources:

1	Microprocessor Architecture, Programming and Applications with the 8085, Ramesh Gaonkar, 2013, Penram International Publishing. .
2	Fundamentals of Microprocessor and Microcomputer, B Ram, 2017, Dhanpat Rai Publications.
3	Advanced Microprocessor and Peripherals, K M Bhurchandi, A K Ray, 2017, McGraw Hill Education.
4	The 8051 Microcontroller, Kenneth J. Ayala, 1996, Penram International Publishing
5	The 8051 Microcontroller and Embedded Systems: Using Assembly and C, M. A. Mazidi, J. G. Mazidi and R D McKinlay, 2007, Pearson.
6.	Microprocessors & Interfacing, Douglas V. Hall and SSSP Rao, 2017, McGraw Hill Education.
7	Computer Organization and Design: The Hardware/Software Interface, David A. Patterson, John L. Hennessy, 2016, Morgan Kaufmann Publishing

Course Name:	Analog and Digital Communication Lab		
Course Code:	PC-EC(V) 491	Category:	Professional Core Courses
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Knowledge of Handling CRO, Function Generator
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05



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Course Objectives:

1	To know the steps involved in the analysis of analog and digital communication systems
2	To generate and observe the waveform of different modulation and demodulation techniques

Course Contents:

Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Measurement of modulation index of an AM signal	2
2	Measurement of output power with varying modulation index an AM signal (for both DSB- &SSB).	2
3	Measurement of power of different frequency components of a frequency modulated signal &the measurement of the bandwidth.	2
4	Design and set up a FM demodulator using PLL	2
5	Measurement of SNR of a RF amplifier	2
6	Measurement of selectivity, sensitivity, fidelity of a super heterodyne receiver.	2
7	Study of Pulse Amplitude Modulation and demodulation	2
8	Study of Pulse Code Modulation and demodulation.	2
9	Study of delta modulator and demodulator.	2
10	Study of adaptive delta modulator and demodulator.	2
11	Study of line coders: polar/unipolar/bipolar NRZ, RZ and Manchester	2
12	Study of ASK, BPSK, BFSK modulator and demodulator.	2
Total		24

Course Outcomes:

After completion of the course, students will be able to:

1	Grow the knowledge about carrier, modulating signal, modulated waveform
2	Apply the concept of VCO in FM generation
3	Apply the concept of signal to noise ratio in noise measurement
4	Acquire skill to analyze and implement analogue to digital converters like PCM, DM, ADM.



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5	Realize different line coding techniques.
6	Understand digital modulation and demodulation technique and interpret the modulated and demodulated waveforms.

Learning Resources:

1	Laboratory Manual
2	B.P. Lathi : Modern Analog and Digital Communication Systems. Oxford.

Course Name:	Python Programming Laboratory		
Course Code:	OE- EC(V)491A	Category:	Open Elective
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Familiar with any Basic computer language
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:

1	a) Master the fundamentals of writing Python scripts b) Learn core Python scripting elements such as variables and flow control structures
2	a) Use Python to read and write files b) Make their code robust by handling errors and exceptions properly
3	a) Explore Python's object-oriented features b) Search text using regular expressions
4	Understand the Exception Handling and Object oriented concept of Python.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction Introduction of Python environment, coding structure. Programming on uses of variables, Data Types, Operators.	2
2	Conditional Statements Programming on If, If- else, Nested if-else, Looping, For, While, Nested loops	2
3	Control Statements Using Break, Continue, Pass statement in Python programming String Manipulation Programming on various operations of String	2
4	Lists/Tuple/Dictionaries Accessing values in Sequence, Working with sequences, Programming on various properties of sequences	3



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5	Functions/Modules Programming on user defined functions and module, Importing module, Math module, Implementation of Array using numpy module,	4
6	Exception Handling Exception, Exception Handling, Except clause, Try- finally clause, User Defined Exceptions	3
7	The Object-Oriented Approach: Classes, Methods, Objects Basic concepts of object programming, Implementation of Properties of Object-oriented Programming, Inheritance.	3
8	File Handling Implementation of various File Operations in programming.	3
9	Data Visualization using Python Concept of Data Visualization, Using Pyplot of Matplotlib library, Creating Line chart, Bar chart and Pie chart using pyplot interface, Customizing the Plot	3
Total		25

Course Outcomes:

After completion of the course, students will be able to:

1	Understand and develop Computational Thinking concepts.
2	Express a problem-solving strategy to breakdown a complex problem into a series of simpler tasks.
3	Describe python programs that appropriately utilize built-in functions and control flow statements.
4	Use functions for structuring Python programs
5	Represent compound data using Python lists, tuples, dictionaries
6	Knowledge of different Charts along with the comparison of different charts.

Learning Resources:

1	"Core Python Programming" by R.Nageswar Rao
2	"Python:the complete reference" by Martin C.Brown
3	"Let us Python" by Yashvant Kanetkar
4	"Programming and Problem Solving With Python" by Ashok Namdev Kamthane and Amit Ashok Kamthane
5	"Python Programming" by Anurag Gupta and G. P. Biswas

Course Name:	Object Oriented Programming laboratory		
Course Code:	OE- EC(V)491B	Category:	Open Elective
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Basic Knowledge of Programming
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05



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Course Objectives:

1	To make the students learn an object oriented way of solving problems
2	To develop their ability to understand a problem and develop solutions using Java as a programming Language.

Course Contents:

Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Assignments on class, constructor, overloading, inheritance, overriding	4
2	Assignments on wrapper class, arrays	4
3	Assignments on developing interfaces- multiple inheritance, extending interfaces	4
4	Assignments on creating and accessing packages	4
5	Assignments on multithreaded programming	4
6	Assignments on AWT and Swing programming	4
7	Assignments on Database Connectivity through JDBC	2
Note: Use Java language for programming		
Total		26

Course Outcomes:

After completion of the course, students will be able to:

1	Use I/O to communicate with the user to populate variables and control program flow
2	Write programs to solve the problems using the features of Object Oriented programming and java technology
3	Develop applications with an event-driven graphical user interface and using databases
4	Practice efficient mapping of the real world interdisciplinary problems into object oriented programming methodologies.

Learning Resources:

1	Patrick Naughton, Herbert Schildt – "The complete reference-Java2" – TMH
2	Deitel and Deitel – "Java How to Program" – 6th Ed. – Pearson
3	Ivor Horton's Beginning Java 2 SDK – Wrox
4	E. Balagurusamy – " Programming With Java: A Primer" – 3rd Ed. – TMH



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Course Name:	Soft Skill Development Lab		
Course Code:	HM-HU491	Category:	Humanities and Social Sciences including Management
Semester:	Fourth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Students must have basic knowledge of English Language.
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	To equip the students with good communication skills.
2	Enable the students to think and speak effectively on everyday topics, including topics related to technical concepts.
3	To prepare them for interviews and future job environments.
4	Developing industry-ready attitude towards professional communication.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Conversation Practice Sessions - General Conversation - Warm-up sessions - Basics of Communication, verbal and non-verbal communication.	4
2.	Group Discussion - Group Discussion & Debates, Do's & Don'ts, etc., Intensive Practice Sessions.	10
3.	Interview sessions: Principles and practices of Personal Interview • Do's and Don'ts of facing an interview. • SWOC Analysis • Rigorous practices of mock-interviews.	6
4.	Presentations: Fundamentals of presentation skills, Secrets of an effective presentation, Presentation Practice Sessions with the help of power point presentation and other audio-visual aids, Face question answer sessions at the end of their presentation.	6
Total		26



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Course Outcomes:

After completion of the course, students will be able to:

1	Honing over all Communicative Competence.
2	Develop Team Building and Leadership Quality.
3	Deliver an enthusiastic and well-practiced presentation
4.	Communicate with clarity and confidence thereby enhancing employability skills of the students.

Learning Resources:

1	Soft Skills: Key to success in Workplace and Life, Meenakshi Raman and Shalini Upadhyay
2	Communication Skills. Sanjay Kumar and PushpLata, Oxford University Press, 2011.
3	Monipally: Business Communication, Tata McGraw Hill
4	Madhukar: Business Communications; Vikas Publishing House
5	Senguin J: Business Communication; Allied Publishers
6.	Business Communication: Rajendrapal & Korlahalli

Course Name:	Constitution of India		
Course Code:	MC472	Category:	Mandatory Courses
Semester:	Fourth	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	Develop an understanding of the nation's constitution.
2	Develop knowledge about the various levels of governance in the country.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction: : Sources and Constitutional history. Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.	4
2	Union Government and its Administration : Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Lok Sabha, Rajya Sabha Supreme Court	6
3	State Government and its Administration Governor.Role and Position, CM and Council of ministers High Court	6



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4	Local Administration District's Administration head: Role and Importance, Municipalities: Introduction, Mayor, and role of Elected Representative. Pachayati raj: Introduction, Zila Pachayat, Elected officials and their roles. Importance of grass root democracy	6
5	Election Commission: Role and Functioning, Chief Election Commissioner	2
Total		24

Course Outcomes:

After completion of the course, students will be able to:

1	Understanding the constitution of India.
2	Understanding the various levels of governance in the country.

Learning Resources:

1	'Indian Polity' by Laxmikanth
2	'Indian Administration' by Subhash Kashyap
3	'Indian Constitution' by D.D. Basu
4	'Indian Administration' by Avasti and Avasti